**Table: 4.2** Bridges classical cross of a triploid (3A+XXX) female fly and a diploid (2A+XY) male fly

| | Triploid ♀ | Diploid ♂ |
|---|---|---|
| Parent | 3A + XXX | 2A + XY |
| Gametes | (2A + XX) (A + X) (2A + X) (A + XX) | (A + X) (A + Y) |

| | A+X | A + Y |
|---|---|---|
| 2A+XX | 3A + XXX Triploid Female | 3A + XXY Triploid Intersex |
| 2A+X | 3A + XX Triploid Intersex | 3A + XY Super Male |
| A+XX | 2A + XXX Super female | 2A + XXY Diploid Female |
| A+X | 2A + XX Diploid Female | 2A + XY Diploid Male |

- X-Chromosome was discovered by Henking (1891)
- Y-Chromosome was discovered by Stevens (1902)

When the X **:** A ratio is 1.00 as in a normal female, or greater than 1.00, the organism is a female. When this ratio is 0.50 as in a normal male or less than 0.50 the organism is a male. At 0.67, the organism is an intersex. metamales (X/A = 0.33) and metafemales (X/A=1.50) are usually very weak and sterile.

A sex–switch gene in *Drosophila* directs female development. This gene, sex–lethal (SxL) located on the X chromosome, has two states of activity. When it is 'on' it directs female development and when it is 'off' maleness ensures. Other genes located on the X chromosome and autosomes regulate this sex-switch gene. However, the Y- chromosome of *Drosophila* is required for male fertility.

## Gynandromorphs

These individuals have parts of their body expressing male characters and other parts of the body expressing female characters. The organism is made up of tissues of male and female genotype and represents a mosaic pattern.

## 4.4.2 Dosage compensation - Barr body

In 1949, Barr and Bertram first observed a condensed body in the nerve cells of female cat which was absent in the male. This condensed body was called sex chromatin by them and was later referred as **Barr body**. In the XY chromosomal system of sex determination, males have only one X chromosome, whereas females have two. A question arises: how

**Table: 4.3 Different doses of X chromosomes and autosome sets and their effect on sex determination in *Drosophila***

| Phenotype | | Number of 'X' Chromosomes (X) | Number of Autosome sets (A) | Sex Index = Number of X chromosome / Number of autosome sets |
|---|---|---|---|---|
| Meta female / Super female | | 3 | 2 | 3/2 = 1.5 |
| Normal Female | Tetraploid | 4 | 4 | 4/4 = 1.0 |
| | Triploid | 3 | 3 | 3/3 = 1.0 |
| | Diploid | 2 | 2 | 2/2 = 1.0 |
| | Haploid | 1 | 1 | 1/1 = 1.0 |
| Inter sex | | 2 | 3 | 2/3 = 0.67 |
| Normal male | | 1 | 2 | ½ = 0.50 |
| Meta male / Super male | | 1 | 3 | 1/3 = 0.33 |



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